

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

df= pd.read_csv("/content/drive/MyDrive/Gen_AI_PBEL/CSV/dataset.csv")
df
```

	Message	Label
0	Hi, how are you?	Ham
1	Let's meet tomorrow.	Ham
2	Call me when you reach home.	Ham
3	Good Morning! Have a nice day.	Ham
4	Can we meet at 5 PM today?	Ham
5	Please send me the assignment.	Ham
6	Happy Birthday! Have a wonderful day.	Ham
7	I'll call you later.	Ham
8	Congratulations! You won a lottery.	Spam
9	Claim your free prize now.	Spam
10	Win cash by clicking this link.	Spam
11	Free recharge available today.	Spam
12	Limited time offer. Click now!	Spam
13	You have won a free iPhone.	Spam
14	Buy one get one free.	Spam
15	Urgent! Claim your reward immediately.	Spam
16	Exclusive offer just for you.	Spam
17	Click here to receive your cash prize.	Spam
18	Your OTP is 456789.	Ham
19	Thank you for your help.	Ham

```
# Input and Output
X = df["Message"]
y = df["Label"]

# Convert text to vectors
tfidf = TfidfVectorizer()
X = tfidf.fit_transform(X).toarray()
```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train KNN
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

# Test Accuracy
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Accuracy: 0.5

```
# User Input
msg = input("Enter a message: ")

# Transform input
msg_vector = tfidf.transform([msg]).toarray()

# Prediction
result = model.predict(msg_vector)

print("Prediction:", result[0])
```

Enter a message: Congratulations! You have won a free iPhone. Click here now!
Prediction: Spam